

NASA Innovative Advanced Concepts (NIAC) — Phase I Step-A Proposal

1. PROJECT IDENTIFICATION & METADATA

- **Proposal Title:** Project Heimdall: Passive Ionospheric Plasma Wake Sensing for Sub-Centimeter Orbital Debris Vetting and Real-Time Trajectory Mapping
- **Principal Investigator (PI):** [Your Name / Institution]
- **Target Mission Directorate:** Space Technology Mission Directorate (STMD)
- **Primary Focus Area:** Space Situational Awareness (SSA), Orbital Safety, and Space Traffic Management (STM)
- **Estimated Technology Readiness Level:** Baseline: TRL 2 | Targeted Phase I Exit: TRL

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2. INNOVATION & CONCEPT SUMMARY

The Critical Capability Gap

Current ground-based Space Surveillance Networks (SSN)—relying on active radar arrays and optical telemetry—suffer from an architectural blind spot. While capable of tracking objects down to approximately 10 cm in Low Earth Orbit (LEO), they are physics-limited against the estimated **1.1 million pieces of anthropogenic space debris between 1 mm and 10 cm**. At hypervelocities ($\sim 7.8 \text{ km/s}$), an impact from a 5 mm fragment delivers the kinetic energy of a ballistic artillery shell, threatening permanent structural failure of human habitats (ISS, Axiom), defense assets, and the catastrophic initiation of Kessler Syndrome cascades.

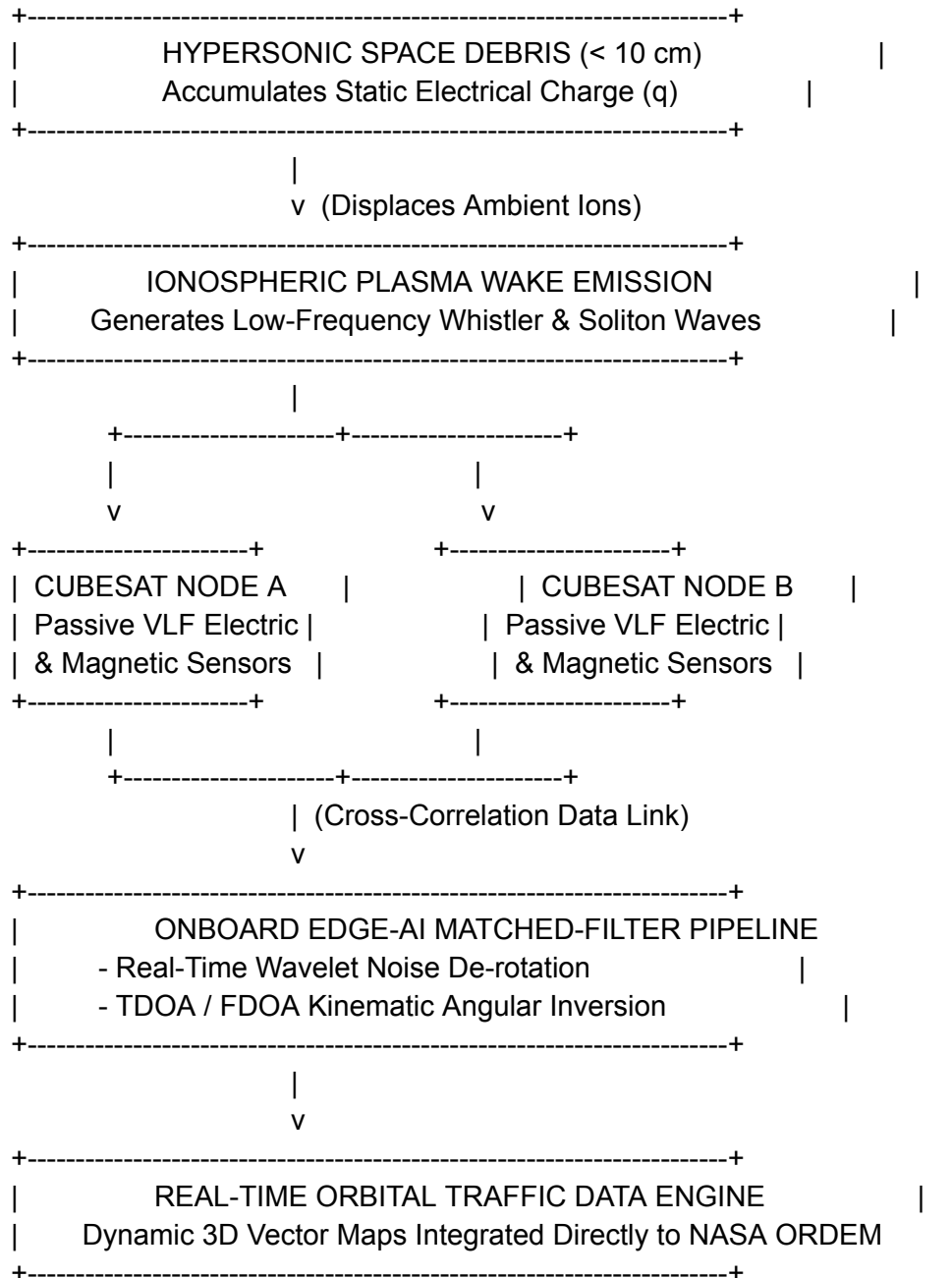
The Breakthrough Concept

Project Heimdall proposes a total paradigm shift: moving away from energy-intensive *active illumination* (radar tracking) and instead utilizing the LEO ionosphere as a natural, macro-scale amplification medium to *passively listen* to space debris.

Objects orbiting in the ionosphere naturally accumulate a net negative static electrical charge (q) via electron collection from ambient plasma. As these hypersonic charged bodies cut across geomagnetic field lines, they fundamentally disturb the local plasma distribution. This creates a hyper-fast, coherent electromagnetic wake characterized by **Whistler-mode waves and Lower Hybrid electrostatic solitons**. Rather than attempting to bounce a signal off a millimeter-sized target, Project Heimdall listens to the massive, loud radio-frequency "shriek" the target naturally radiates into the ionospheric medium.

3. TECHNICAL ARCHITECTURE

Project Heimdall utilizes a dual-element architecture split precisely between sparse, low-power orbital hardware sensors and a distributed, edge-computing software framework.



Element A: Hardware Sensor Network (The Observers)

The hardware footprint relies on a constellation of 3U CubeSats deployed in sun-synchronous orbits. Each platform requires minimal power ($\leq 5 \text{ W}$) because it functions exclusively as a passive listening station.

1. **Antenna Array:** A deployable orthogonal tri-axial electric dipole antenna combined with an induction search-coil magnetometer.

2. **Telemetry Profiling:** The node continuously samples voltage fluctuations and magnetic fields across the Very Low Frequency (VLF) bands (10 Hz to 30 kHz), transforming the physical medium into a highly continuous digit stream.

Element B: Deep-Software Signal Processing Pipeline (The Brain)

The foundational intellectual breakthrough lies in the software processing pipeline, optimized to run on radiation-hardened edge processors:

1. **Dynamic Noise Subtraction:** The ionosphere is heavily saturated with background noise (auroral signals, lightning-induced whistle noise). The algorithm applies localized, real-time spatial filtering to strip this ambient noise from the telemetry feed.
2. **Wavelet Matched-Filtering:** When a debris fragment cuts through the detection zone, its hypersonic wake produces a distinct, mathematically predictable dispersion curve. A custom machine learning model runs constant mathematical cross-correlations to instantly isolate this exact signature.
3. **Kinematic Triangulation:** By measuring the **Time Difference of Arrival (TDOA)** and **Frequency Difference of Arrival (FDOA / Doppler Shift)** across multiple neighboring CubeSat nodes, the software dynamically solves the inverse kinematic equations to isolate the object's exact velocity vector ($\vec{v}$) and inclination (i).

4. MISSION CONTEXT & SYSTEM FEASIBILITY

Within the context of NASA's long-term operations, Project Heimdall integrates seamlessly as a plug-and-play data stream for the **Orbital Debris Program Office**.

Instead of waiting for an asset to get hit, Heimdall provides a real-time, high-density 3D traffic map of sub-centimeter debris "clouds." Commercial crew vehicles and robotic deep-space science platforms can utilize this prognostic data engine to map out optimized, low-risk trajectories through LEO before ever initiating a launch sequence.

5. PHASE I RESEARCH OBJECTIVES

The 9-month Phase I program will focus strictly on validating the foundational physics and software viability:

- **Task 1 (Physics Simulation):** Develop high-fidelity Particle-in-Cell (PIC) computational plasma simulations to accurately characterize the wave modes generated by varied debris geometries (metallic shards, carbon-composite flakes, paint chips).
- **Task 2 (Algorithm Benchmarking):** Program the Wavelet Matched-Filter framework and evaluate its detection sensitivity against synthetic data injected with realistic ionospheric signal-to-noise ratios (SNR).
- **Task 3 (Feasibility Assessment):** Define the minimum baseline constellation spacing required to ensure 95% global detection coverage for objects under 1 cm in high-value orbital corridors.

